

Raidium Challenge 2025

CT Scan Organ Segmentation

Vaishali Vinod

Karolina Saegner

Applied Machine Learning

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Introduction

The Task

Automatically segment anatomical structures from 2D CT scan slices

Image segmentation — pixel clustering task

Metric: Mean DICE score

$2|A \cap B| / (|A| + |B|)$, nanmean over classes then images

Why It Is Hard

- No consistent label mapping across images
- Many organs share similar CT intensity
- Thin structures collapse at low resolution
- No radiological pretrained models allowed

Literature Review

Watershed

Meyer, 1994

Classical gradient-based segmentation. Parameter-free. Used as our baseline, the original algorithm underpins our baseline model.

Dice Loss

Milletari et al., 2016

Soft Dice directly optimises the evaluation metric. Combined with CrossEntropy for stable training. Applied in our best model.

U-Net

Ronneberger et al., 2015

Encoder-decoder with skip connections. De facto standard for medical image segmentation. Directly inspired our architecture.

SMP

Iakubovskii, 2019 — `segmentation_models_pytorch`

Python library providing ready-made segmentation architectures with pre-trained encoders. Used throughout this project to build all four final models.

Dataset Overview

2,000

Training Images

500

Test Images

256×256

Resolution (px)

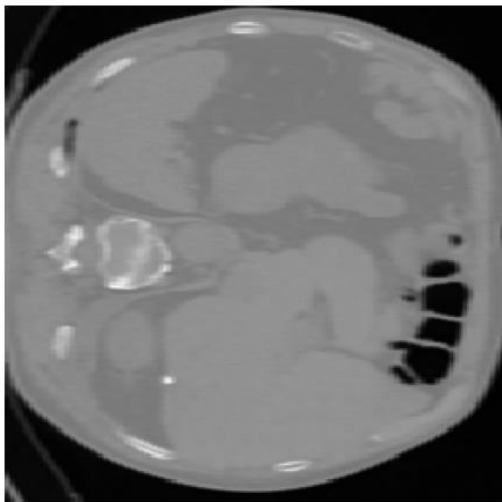
54

Organ Classes

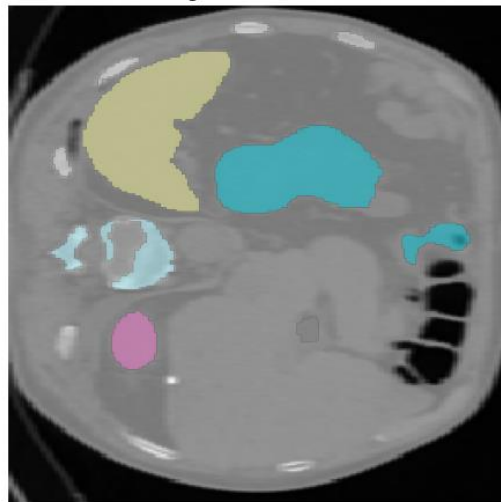
Sample #150

Sample #50

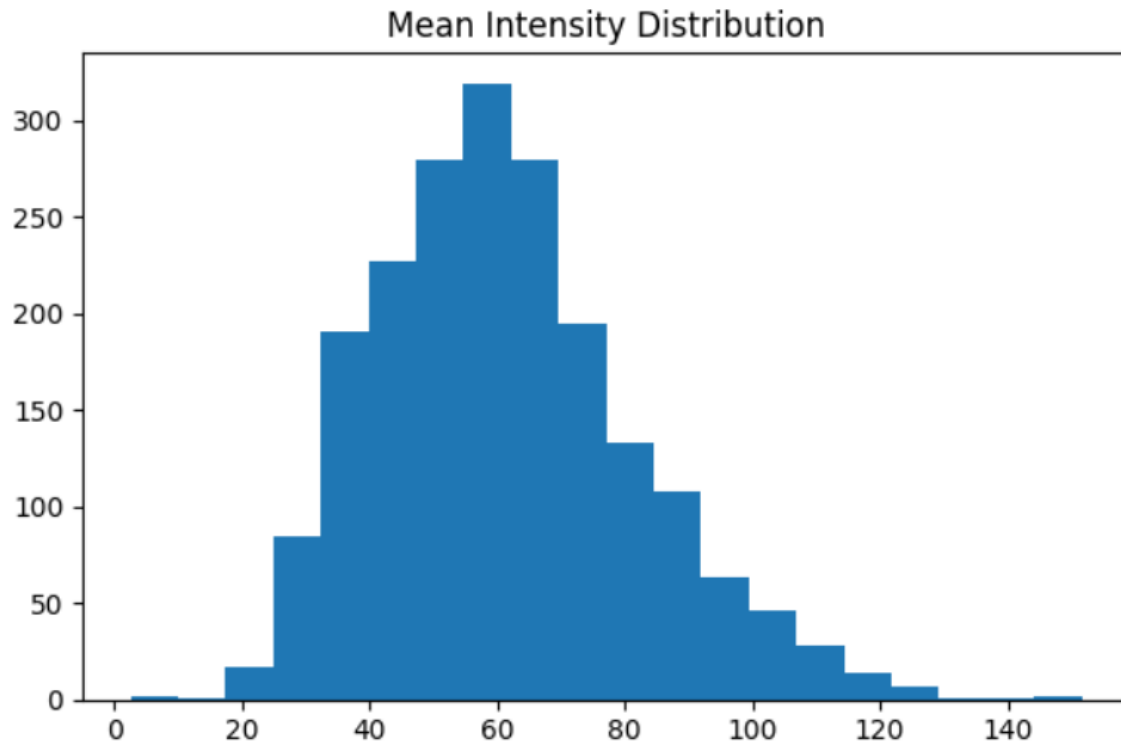
CT Slice



Segmentation Mask



Grayscale Intensity Analysis



Intensity skews heavily toward low values — expected CT behaviour.

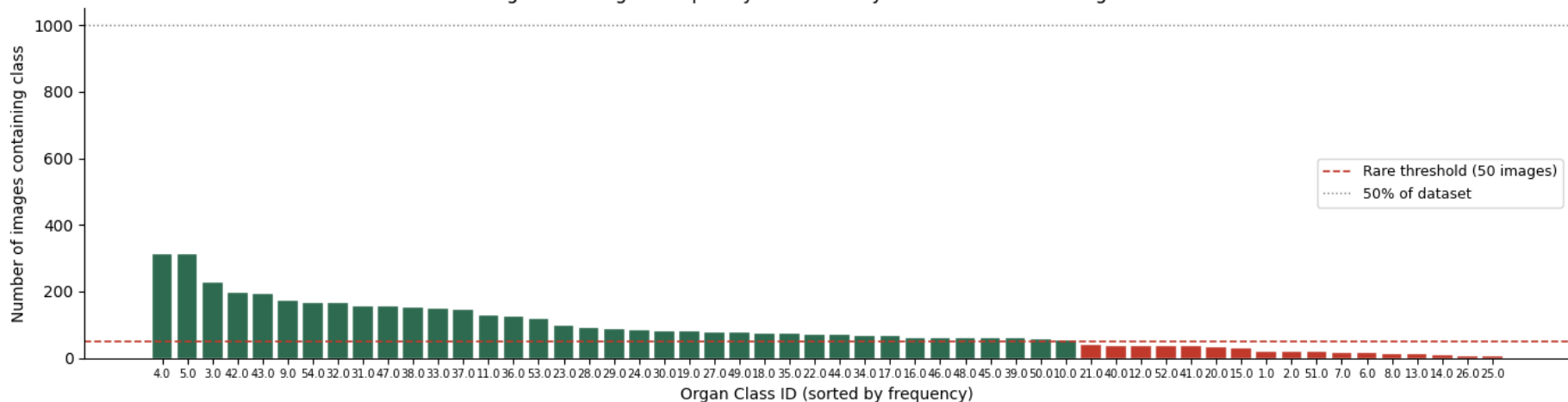
Organ labels

Binary matrix shape (images × organs): (2000, 54)

	1	2	3	4	5	6	7	8	9	10	...	45	46	47	48	49	50	51	52	53	54
0	0	1	0	1	0	0	0	1	0	1	...	0	1	1	0	1	1	0	1	1	1
1	0	0	0	1	0	1	0	0	0	1	...	1	1	1	1	1	0	1	0	1	1
2	0	1	1	0	0	1	1	1	1	0	...	0	0	0	0	1	1	0	0	0	1
3	1	0	0	0	0	1	1	0	0	0	...	0	0	1	1	0	0	1	1	1	1
4	0	0	0	1	1	0	0	0	1	0	...	1	0	0	1	0	1	0	1	0	1

Organ Frequency

Image-Level Organ Frequency — how many scans contain each organ class



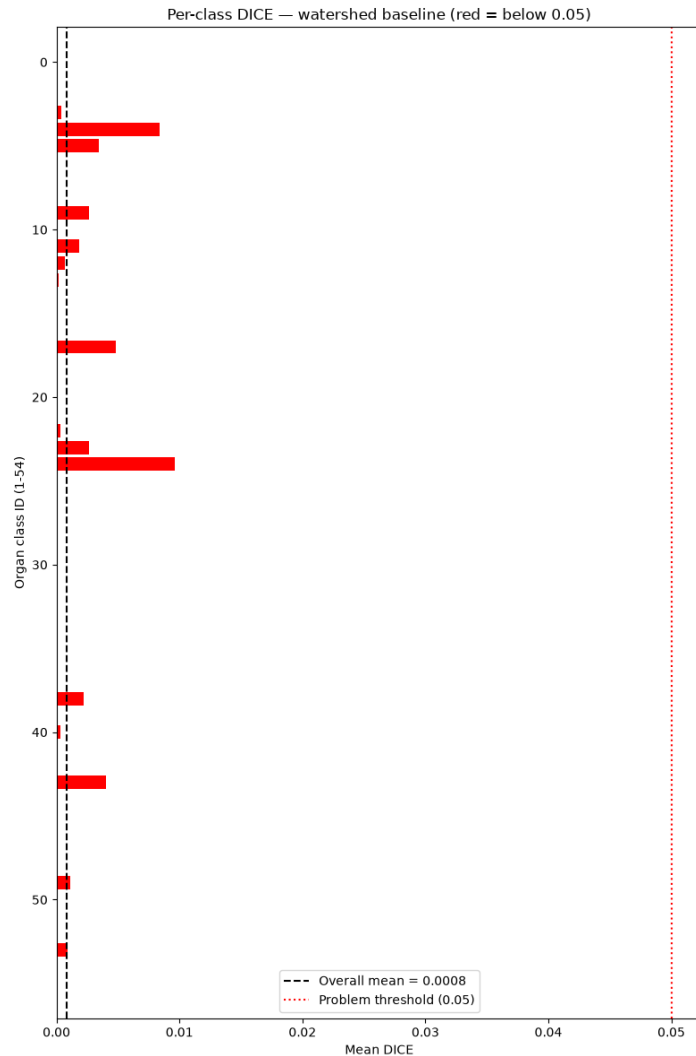
- Mean 22 organs per image
- Range 1–54

Baseline Model: Watershed Segmentation

Why?

- Standard starting point in medical image segmentation literature
- No training needed
- Produces a meaningful lower bound
- DICE near zero proves semantic learning is essential
- Sets the floor every supervised model must beat

Overall DICE: **0.0008** → lower bound for all supervised models



Baseline Model: Watershed Segmentation

5 Zero-DICE Classes

Class	DICE	Why it fails
Class 1	0.0000	Absorbed into large background region
Class 2	0.0000	Near-zero gradient vs surroundings
Class 6	0.0000	Thin tubular structure — over-segmented
Class 8	0.0000	Fewer than 20 pixels; smoothed away
Class 14	0.0000	Ring lumen splits into two wrong fragments

Overall DICE: **0.0008** → lower bound for all supervised models

Training Setup & Hyperparameters

Data Pipeline

- 1 Load 256×256 PNG → numpy (N×H×W)
- 2 Reshape flat CSV labels → (N, 256, 256) int64 masks
- 3 Val split: first 200 images held out
- 4 CTSegDatasetRGB: normalise [0,1] · replicate grayscale to 3 channels · ImageNet mean/std normalisation
- 5 Augment (train only): hflip · vflip · brightness ±20%
- 6 DataLoader: batch=8 · pin_memory=True · 2 workers

Hyperparameters

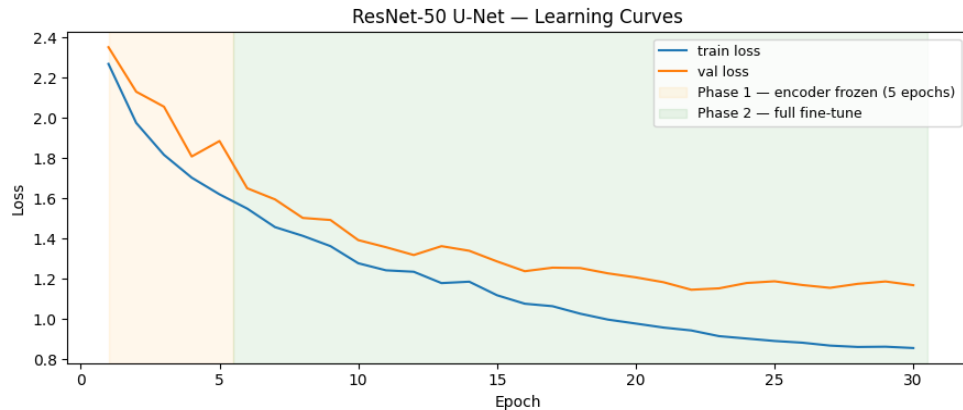
Batch size	8 (T4 safe at 256 ²)
Epochs	30
Scheduler	Cosine Annealing
Weight decay	1e-4
Precision	AMP fp16 (auto)
Phase 1	encoder frozen, decoder trains at LR = 3e-4
Phase 2	encoder unfrozen at LR = 3e-5, decoder at LR = 3e-4
Val set	200 images (fixed)

Model #1

ResNet-50 U-Net

- Pre-trained ResNet-50 encoder (ImageNet)
- U-Net decoder with skip connections
- ~32 M params total · encoder ~23 M
- Class-weighted CE + Dice loss
- Two-phase training: encoder frozen 5 epochs, then differential LR (encoder 10× lower)

~15 min on T4

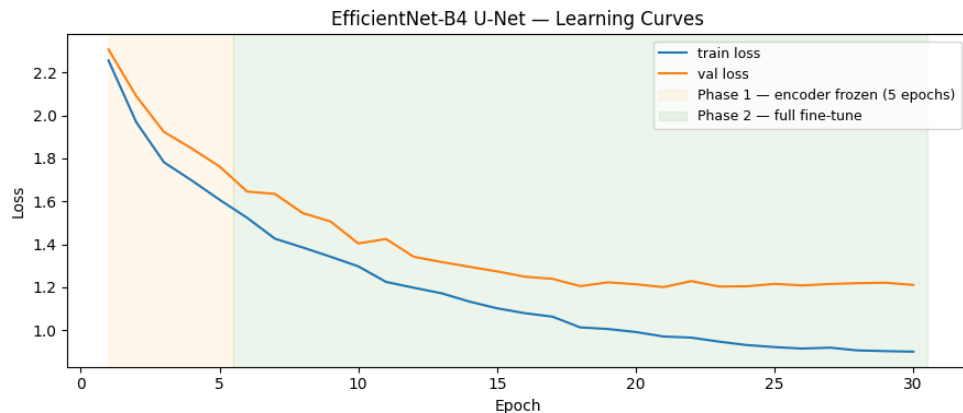


Model #2

EfficientNet-B4 U-Net

- Pre-trained EfficientNet-B4 encoder (ImageNet)
- U-Net decoder with skip connections
- ~24 M params total · encoder ~17 M
- Compound-scaled CNN: better accuracy/param ratio than ResNet
- Class-weighted CE + Dice loss

~15 min on T4

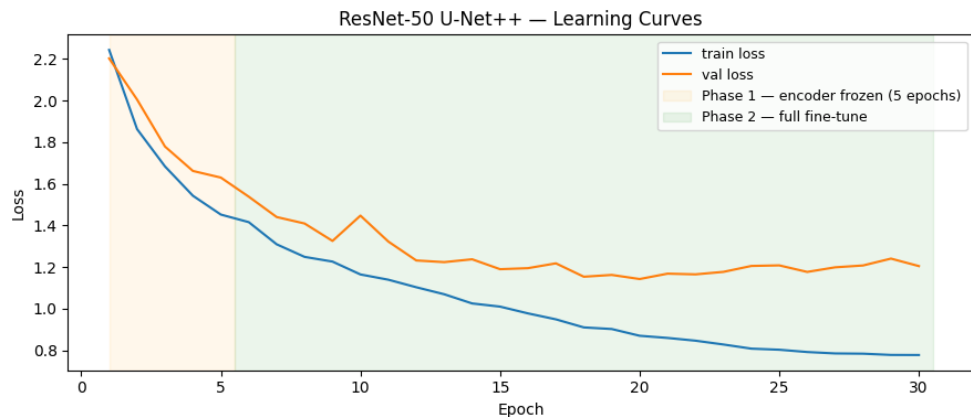


Model #3

ResNet-50 U-Net++

- Pre-trained ResNet-50 encoder (ImageNet)
- U-Net++ decoder: dense nested skip connections
- ~55 M params total · encoder ~23 M
- Same encoder as #1 — isolates the decoder impact
- Class-weighted CE + Dice loss

~18 min on T4

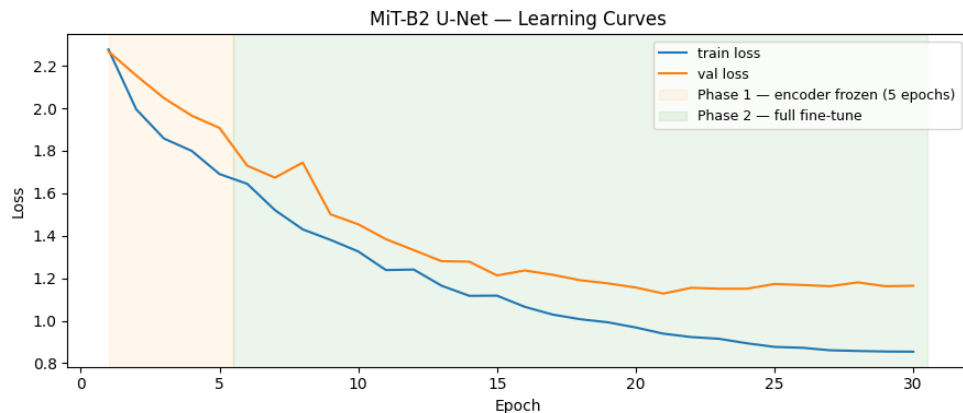


Model #4

MiT B2 U-Net

- Pre-trained Mix Transformer B2 encoder (ImageNet)
- U-Net decoder with skip connections
- ~28 M params total · encoder ~25 M
- Attention-based encoder captures long-range spatial context
- Helps thin/tubular structures missed by CNN encoder

~18 min on T4



Model Evaluation

ResNet-50 U-Net

32.5M params

Encoder: 23.5M

ResNet-50 ImageNet
encoder

U-Net decoder with skip
connections

Phase 1: encoder frozen 5
epochs

Phase 2: full fine-tune 25
epochs

DICE: 0.1122

102× baseline

EffNet-B4 U-Net

20.2M params

Encoder: 17.5M

EfficientNet-B4 ImageNet
encoder

Smaller but more efficient
CNN

Same two-phase training
strategy

Lightest model in the
comparison

DICE: 0.0934

85× baseline

ResNet-50 U-Net++

49.0M params

Encoder: 23.5M

Same ResNet-50 encoder
as Model 1

U-Net++ decoder: dense
nested skips

Isolates the decoder
improvement

Most parameters of all 4
models

DICE: 0.1194

109× baseline

MiT-B2 U-Net

27.5M params

Encoder: 24.2M

Mix Transformer B2
encoder

Attention captures long-
range context

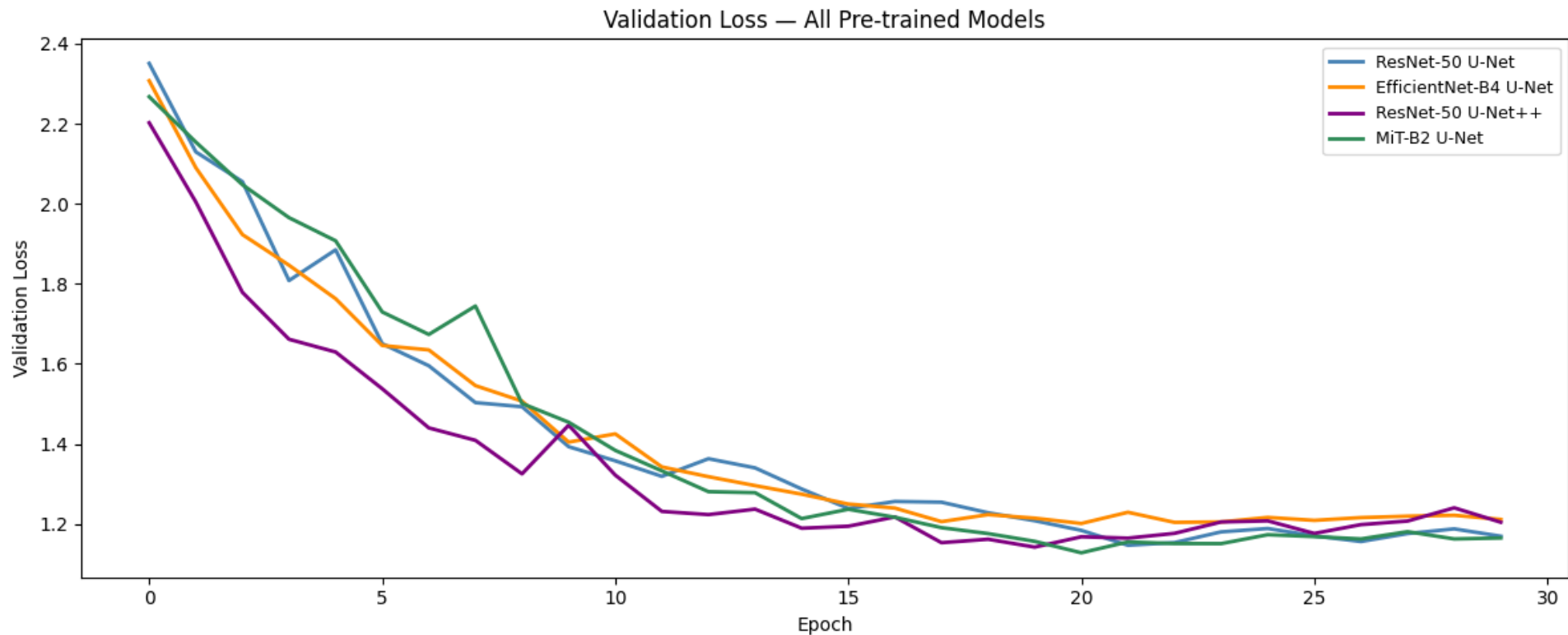
Helps thin/tubular organ
structures

Best overall result in this
run

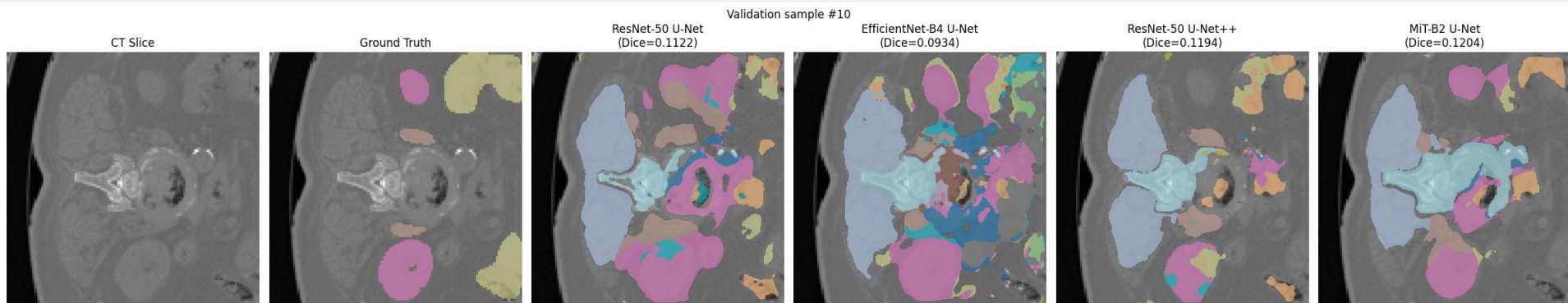
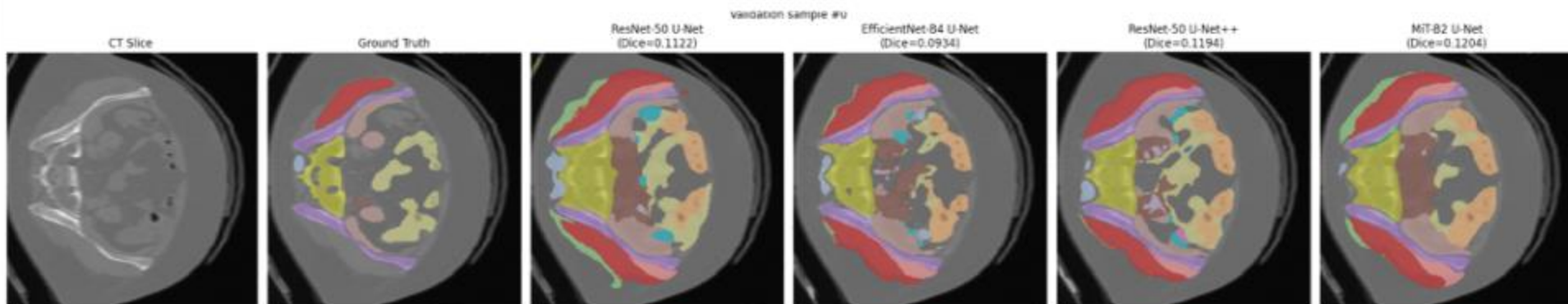
DICE: 0.1204

109× baseline ✓

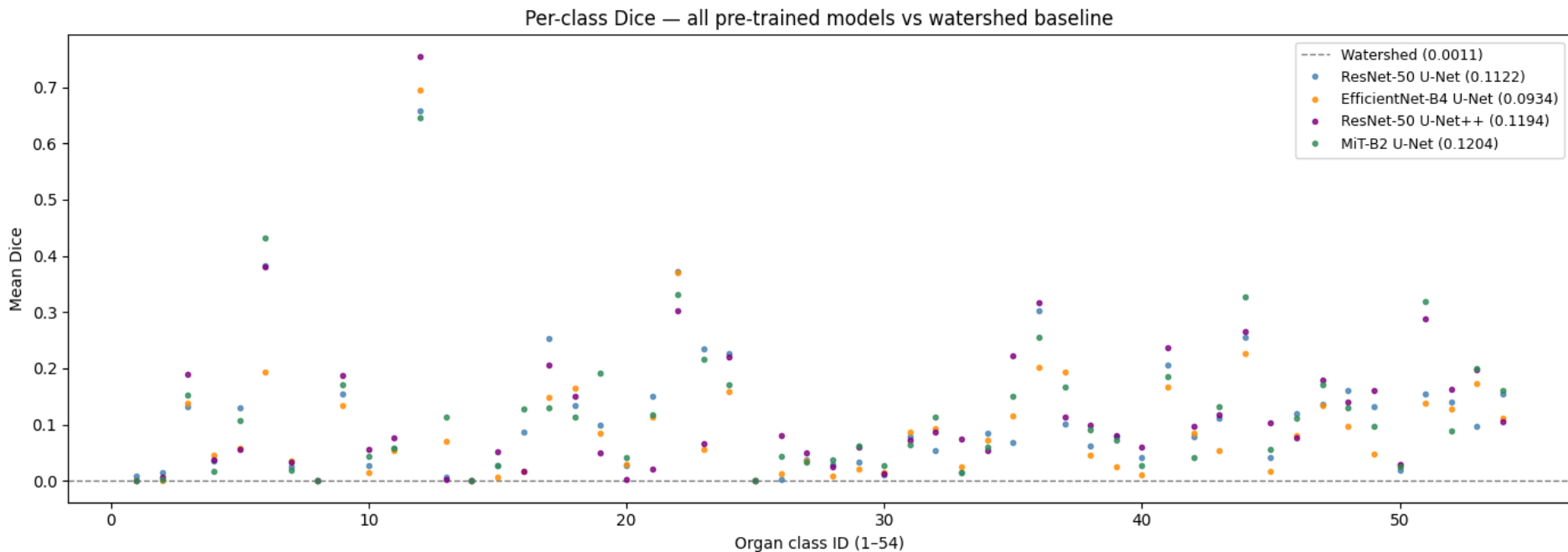
Model Evaluation



Model Evaluation



Per-class DICE — All Models vs. Watershed



All four pre-trained models outperformed the watershed baseline, with scores between 85× and 109× higher. The transformer-based MiT-B2 came out on top.

Summary & Next Steps

What We Found

- All 4 pre-trained models exceeded the watershed baseline by $\sim 100\times$
- MiT-B2 U-Net scored highest (DICE 0.1204), ResNet-50 U-Net++ close behind (0.1194)
- U-Net++ decoder gave +0.007 over U-Net with the same ResNet-50 encoder
- EfficientNet-B4 underperformed (0.0934) — compound scaling transfers less well to medical images
- Two-phase training worked reliably across all four models

Next Steps

1

Ensemble MiT-B2 + ResNet-50 U-Net++, similar scores suggest complementary errors

2

Try MiT-B2 + U-Net++ decoder — best encoder and best decoder not yet combined

3

Test-Time Augmentation (TTA) — free $\sim 1\text{--}3\%$ DICE gain at inference

4

Train at 512×512 — thin structures collapse at 256px

Best result: MiT-B2 U-Net · DICE 0.1204 · $109\times$ watershed baseline

Conclusion and take away

- Transfer learning matters most
- Class weighting was the most impactful training decision, without which rare organs were invisible to the loss function
- More parameters $\neq$ better results — EfficientNet at 20M params underperformed ResNet at 32M, and U-Net++ at 49M only marginally beat ResNet U-Net at 32M
- The dataset is the hardest part — inconsistent labels across images, class imbalance, and thin structures were bigger obstacles than model architecture
- 0.12 DICE is real progress but not clinical — the gap to clinical acceptance (>0.7) shows how hard medical segmentation actually is